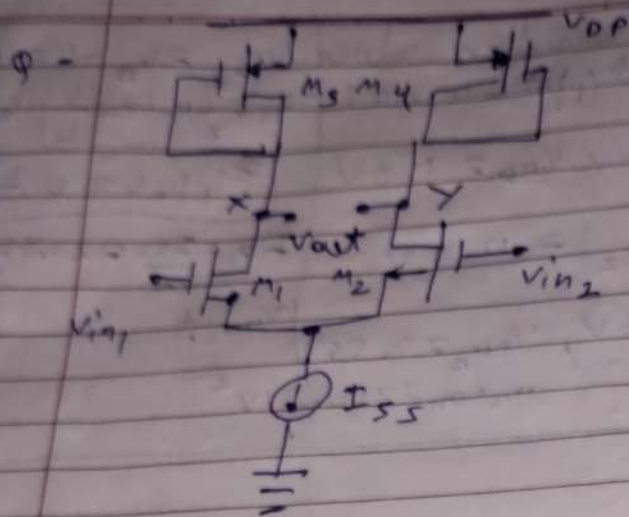
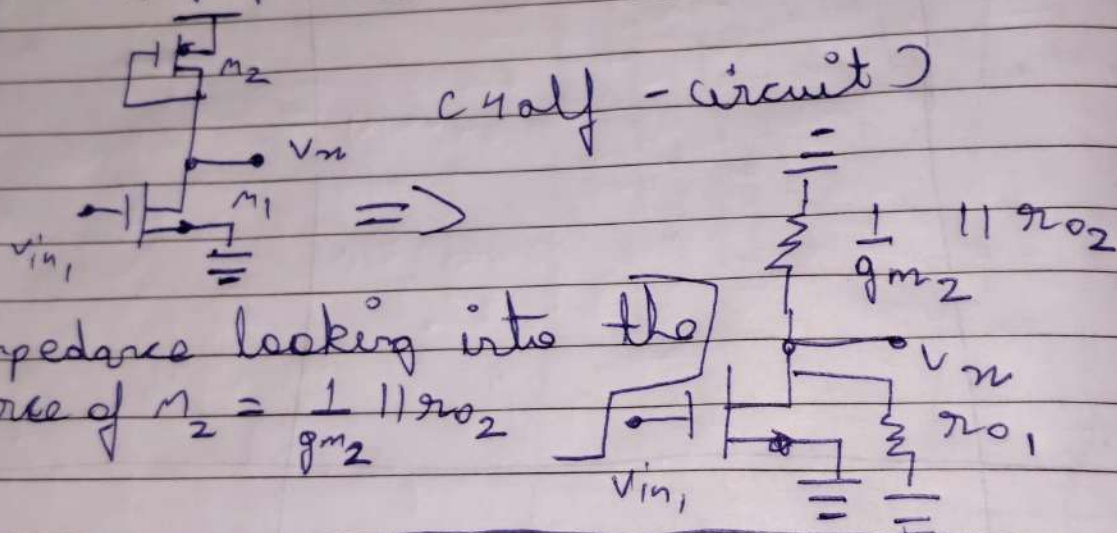




Values :-
 $V_{DD} = 1.8V$, $\frac{W}{L} = 1$
 $V_{THN} \approx 0.65V$
 $V_{THP} \approx -0.6V$
 $\mu_n = 500 - 600 \text{ cm}^2/\text{Vs}$
 $\mu_p = 200 \text{ cm}^2/\text{Vs}$
 $C_{ox} = 8.4 \text{ fF}/\mu\text{m}^2$
 $\theta_P = 92 \text{ mV/V}$
 $\theta_n = 145 \text{ mV/V}$

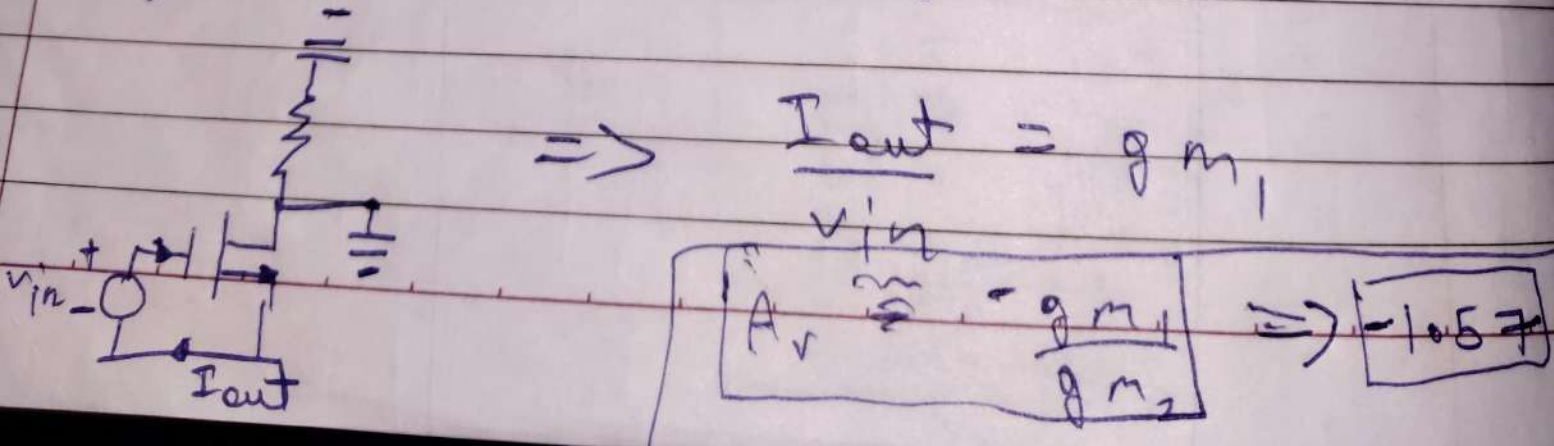
Small-Signal Analysis :- (Assuming Symmetry)
 (Applying Differential signal)



Impedance looking into the source of $M_2 = \frac{1}{g_{m2}} \parallel r_{o2}$

$$R_{out, total} = \frac{1}{g_{m2}} \parallel r_{o2} \parallel r_{o1}$$

Approx G_m Calculation :-



We can ground V_P as it acts as a virtual ground when we apply differential signal.

$$g_{m2} = \sqrt{2 \times \mu \times C_{ox} \times \frac{W}{L} \times I_D}$$

$$g_{mn} = \sqrt{2 \times (420 \times 10^{-6}) \times 6.67 \times 3.75 \times 10^{-6}} = \sqrt{2.1 \times 10^{-8}} \approx \underline{\underline{145 \mu A/V}}$$

$$g_{mp} = \sqrt{2 \times (160 \times 10^{-6}) \times 6.67 \times 3.75 \times 10^{-6}} = \sqrt{8.4 \times 10^{-9}} \approx \underline{\underline{92 \mu A/V}}$$